

# Protocols are Systems

Implications for a Grand Unified Protocol Modeling Theory



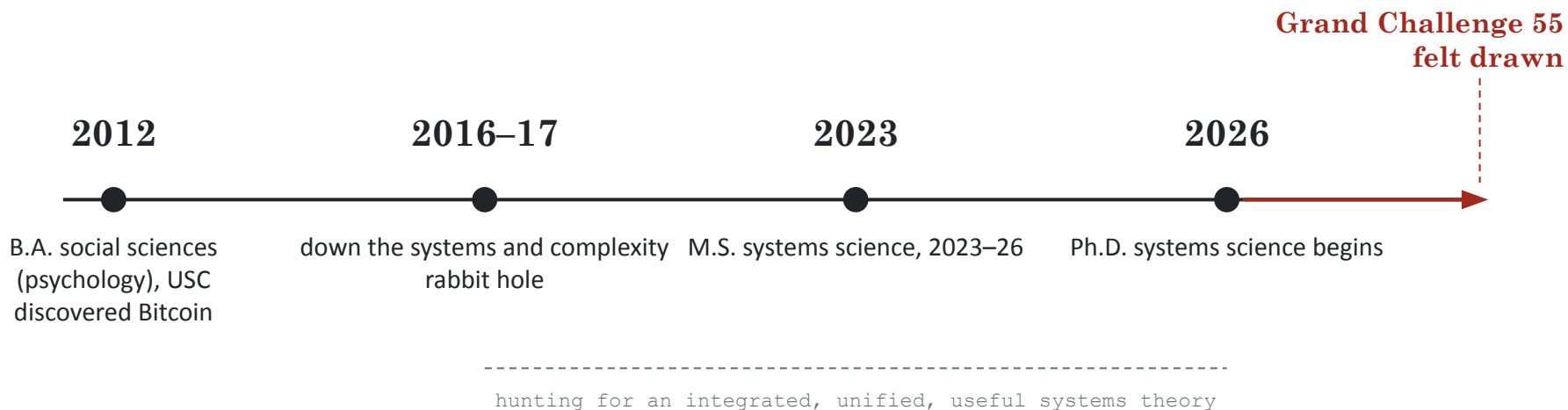
# Grand Challenge 55

## a Grand Unified Protocol Modeling Theory

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1. Explore **existing formalisms** for **modeling systems** that could be described as protocols (state machines, process calculi, temporal logics, etc.)
2. **Unify them elegantly** in a way that captures the observed behavior of most protocols.
3. Develop a non-brittle, **evolvable** modeling formalism that enables:
  - a. Future research
  - b. **Proving of results**
  - c. **Accommodation** of the phenomenology of **new protocols**

# Why I'm here



# What is a protocol?

**Rules** created by humans that shape the behavior of **groups of people**.

Walch 2023

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A simple **codified set of behaviors** that, when adopted by a sufficient number of **participants**, reliably leads to good enough outcomes for all.

Rao, Beiko, Ryan, Stark, Van Epps, Aue 2023

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Explicit or implicit **rules or procedures** intended to govern, guide, or influence **behaviour** in a way that allows the emergence or construction of coordinated behavior.

Tay 2024

things

relations

# What is a system?

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## CONSTRUCTIVIST

Klir

Things, and relations between them.

$$S = (T, R)$$

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## REALIST

Bunge

Composition, environment, structure, mechanism.

$$S = \langle C, E, S, M \rangle$$

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## PERSPECTIVIST

Mobus

Components, internal networks, environment, external flows, boundary, transformations, history, time scale.

$$S = \langle C, N, E, G, B, T, H, \Delta t \rangle$$

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# Protocols are systems



**T, things** = the participants. People, agents, machines.

**R, relations** = the rules. Codified, adopted, constraining what participants do.

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A protocol is (T, R). That is Klir's definition of a system.

# Hand-washing as things and relations

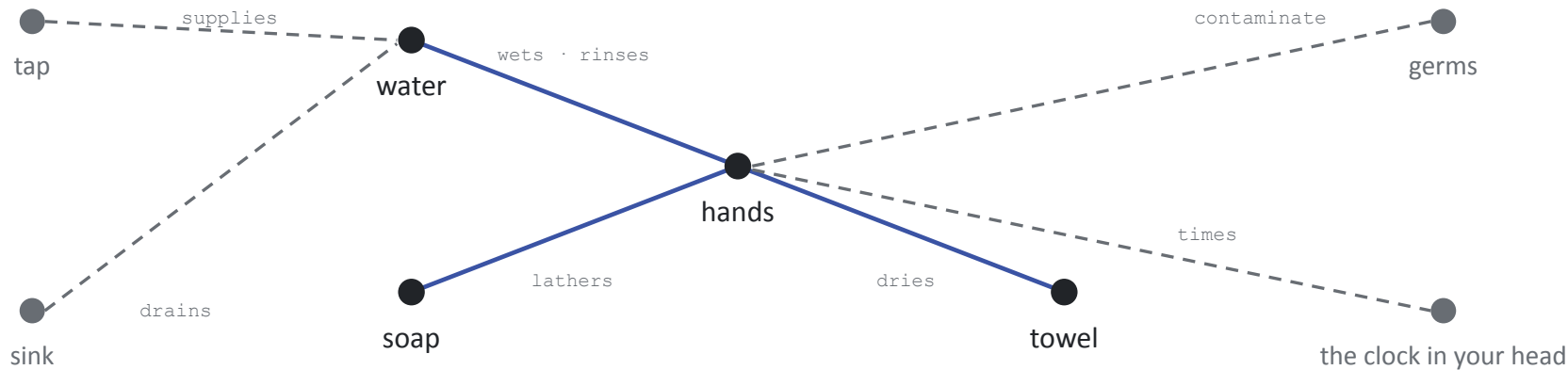
$S = (T, R)$

| T, things | R, relations                |
|-----------|-----------------------------|
| hands     | (water, hands) wets         |
| water     | (soap, hands) lathers       |
| soap      | (hands, hands) scrubs, 20 s |
| towel     | (water, hands) rinses       |
|           | (towel, hands) dries        |

T is a table of entities. R is a table of pairs. That is all Klir asks for, and it is a data scientist's view of a system.

# Hand-washing as a real thing

$S = \langle C, E, S, M \rangle$

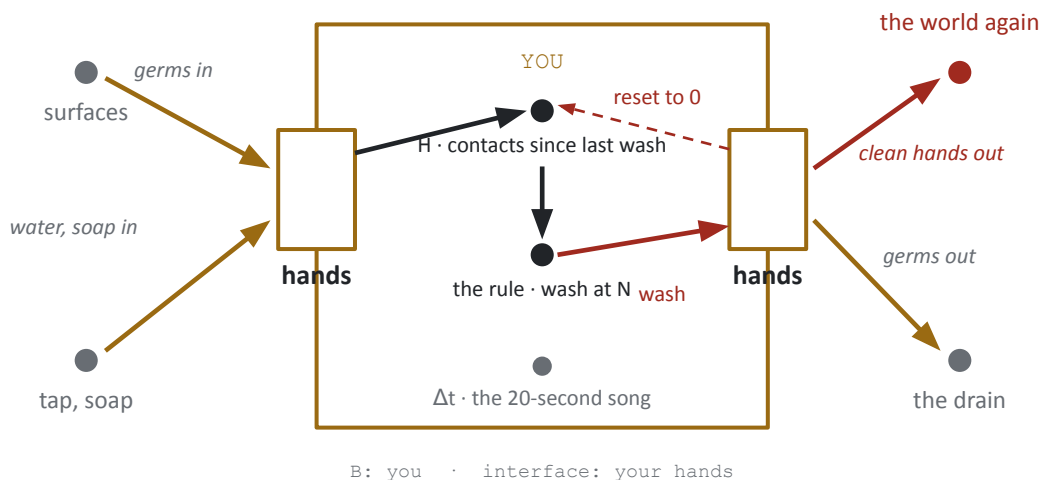


black: C, the composition · grey: E, the environment · lines: S, the structure

M, the mechanism: lather lifts germs off skin, water carries them to the drain, drying keeps them off.

# Hand-washing as dynamic process

$$S = \langle C, N, E, G, B, T, H, \Delta t \rangle$$

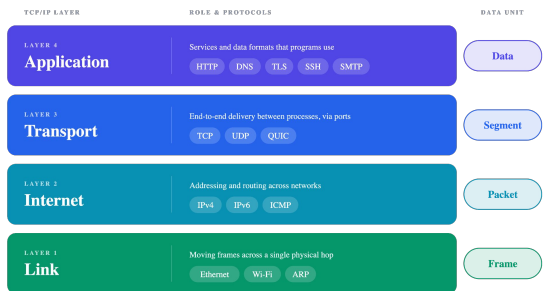


- C** your hands, and the rule you follow
- N** each touch counts; at N the rule sends you to wash
- E** surfaces, the tap and soap, the drain, the world
- G** germs, water, soap in; germs out; clean hands out
- B** you; your hands are the interface
- T** contaminated hands in, clean hands out
- H** contacts since the last wash
- $\Delta t$**  20 seconds to scrub, one Happy Birthday

**Output and purpose live in G and T.** "Clean hands" is the good-enough outcome.

# Protocols that run the world

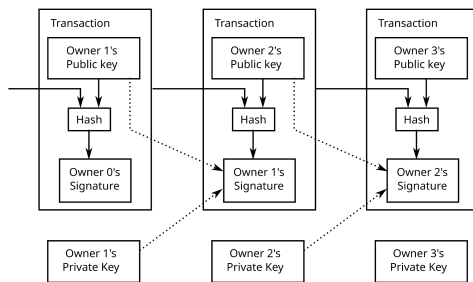
Technical, embedded in the fabric of society, amenable to formal analysis.



## TCP/IP

things hosts, packets

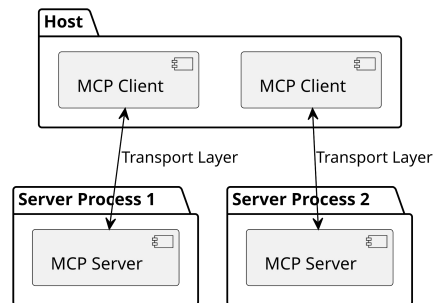
relations handshake,  
acknowledgment, retransmit



## Bitcoin

things miners, nodes, the ledger

relations consensus rules, difficulty  
adjustment



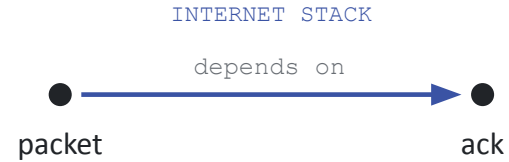
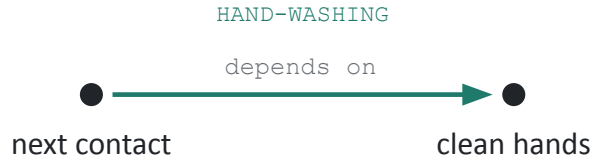
## MCP

things agents, tools, servers

relations the tool-call exchange

# The common core

Only the blanks change.



A system is things that depend on other things. Every definition on slide 5 keeps this arrow.

# Protocols are systems. So what?

What a system model buys, against Venkat's challenge.

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1      **Unify the existing formalisms**

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2      Evolvable, not brittle

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3      Proving of results

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4      Accommodation of new protocols

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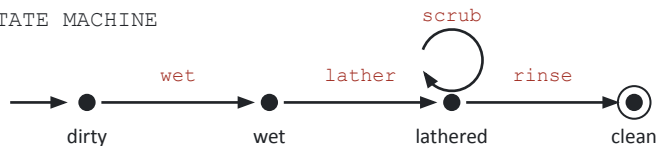
5      Future research

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# One skeleton, many formalisms

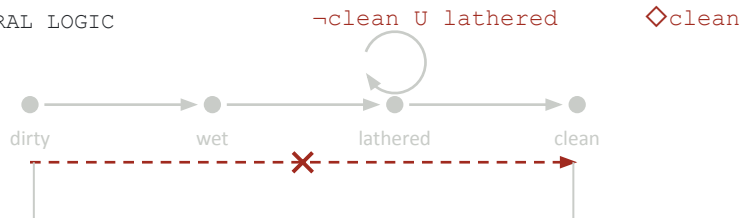
Every protocol formalism draws the same picture: things, and the relations between them.

STATE MACHINE



names each step

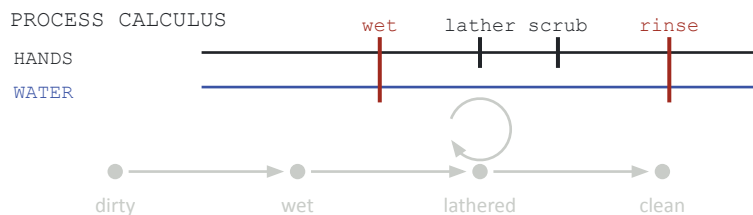
TEMPORAL LOGIC



rules over the whole run: no clean without soap

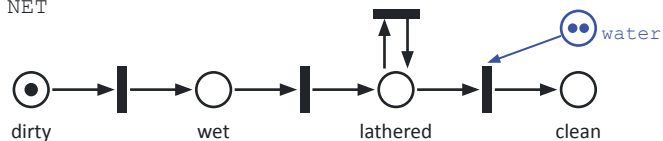
the same four states in all four

PROCESS CALCULUS



two players, meeting at wet and rinse

PETRI NET

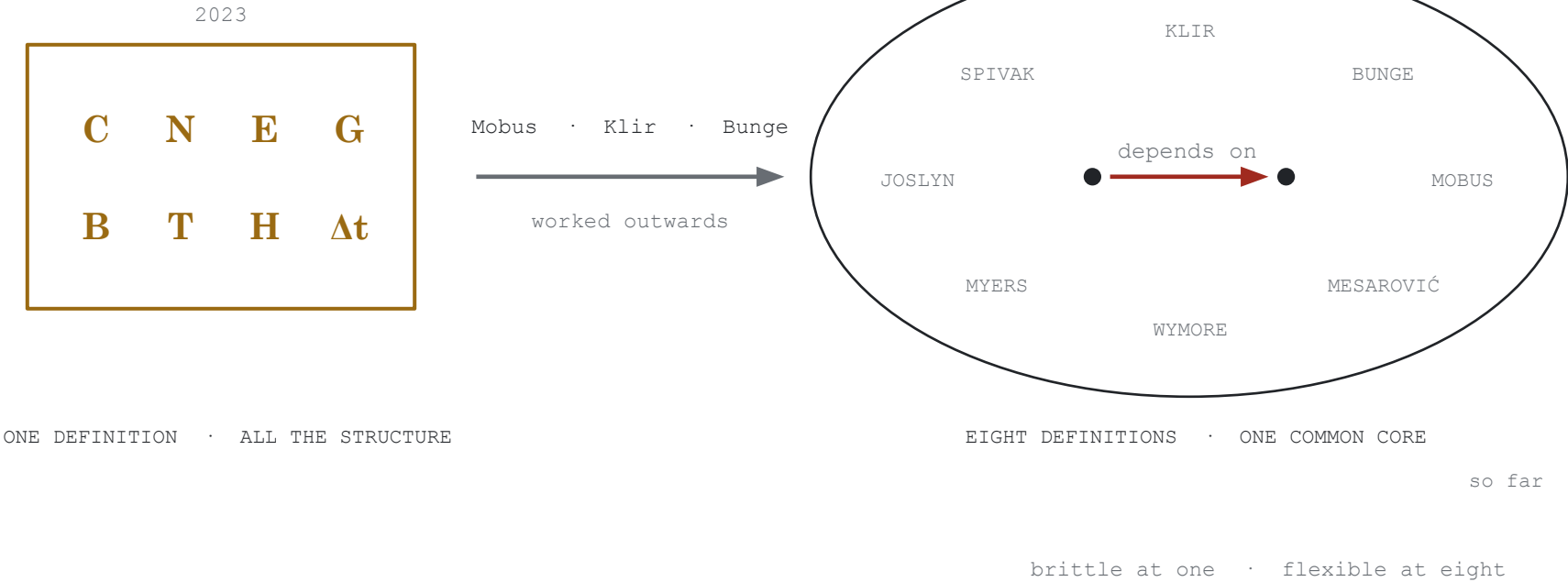


it can count: here, the water

a sketch, not a result

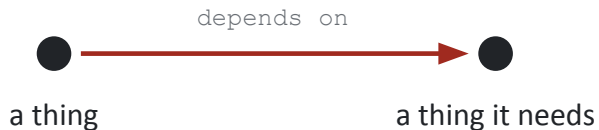
# Evolvable, not brittle

Plural integration beats a theory of everything.



# Proving of results

Three protocols, three theorems, one arrow. Checked by Lean, not by me.



HAND-WASHING

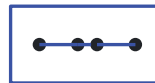
`handwashing_contamination_bounded`



the count never exceeds N

TCP + HTTP

`stack_organized`



joined by one bond, still a system

BITCOIN

`difficulty_target_is_equilibrium`



10 minutes is a fixed point

# What the checker read

The comment is the claim in English. The next line is the claim in Lean.

Protocols/HandWashing.lean · 269-275

```
-- The safety invariant: starting anywhere in the viable region (contamination  
at most the period), the every-N rule keeps contamination at most the period  
for any number of steps. The viable set is forward-invariant. -/
```

```
theorem handwashing_contamination_bounded (n c : ℕ)  
  (hsafe : c ≤ sanitizePeriod) :  
  contaminationStep^[n] c ≤ sanitizePeriod
```

Protocols/TCP/IP.lean · 118-124

```
-- Two protocol layers, composed, yield again a valid organized system,  
with no compatibility precondition. -/
```

```
theorem stack_organized : IsOrganized stack.composition
```

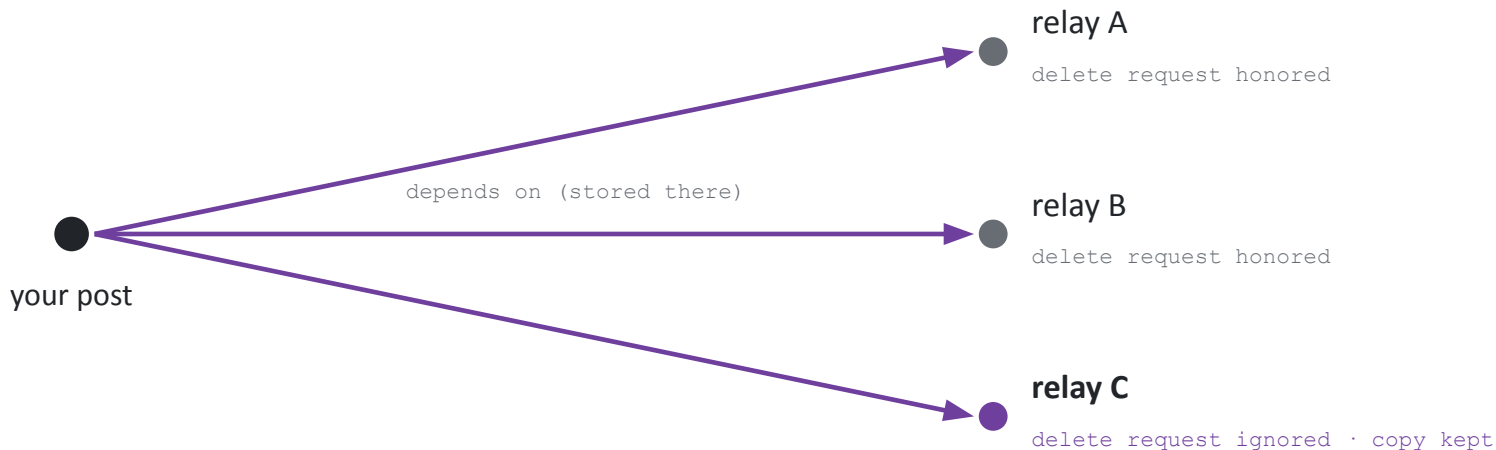
Protocols/Bitcoin.lean · 236-241

```
-- The 10-minute target block interval is a fixed point of the difficulty  
feedback law: once the inter-block time is at target, the retarget rule  
leaves it there. -/
```

```
theorem difficulty_target_is_equilibrium :  
  IsEquilibrium difficultyHomeostat.feedbackLaw 10
```

# Accommodation of new protocols

Publishing is one signed message. Deleting is a request any relay can ignore.



# Future research

Three directions worth pursuing.

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1 Can hardness be measured, not just located?

toward scales and metrics that work across protocols

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2 Does the unification hold past four formalisms?

encode a fifth; see if the skeleton survives

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3 Where does this model break on a protocol you care about?

where the models break may be the most valuable result

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# The ask: protocol analysis, literally

Send a spec, answer a short survey, let me interview you.

1 · SPEC

a link to your protocol's formal  
spec, if one exists



2 · SURVEY

role · tenure · self-assessed  
understanding



3 · INTERVIEW

a conversation about your  
protocol

follow-up in the FPT Discord

# What does the protocol you care about look like as a system?



continuing in the FPT Discord

# Receipts

Lean 4. Three instance files: `Protocols/HandWashing.lean` · `Protocols/TCPIP.lean` · `Protocols/Bitcoin.lean`

Theorems: `handwashing_contamination_bounded` · `stack_organized` · `difficulty_target_is_equilibrium`

The eight definitions: `Klir` · `Bunge` · `Mobus` · `Mesarović` · `Wymore` · `Myers` · `Joslyn` · `Spivak`  
(systems-science-foundations, eight `klirTo*_faithful` theorems)

All instance files build green, zero sorry, no axioms beyond Lean's standard three (verified 2026-08-05)

Research support: C3PO for corpus navigation · Claude + Lean for formalization · files and build available on request